CHORD EXTENSION NUMBERS

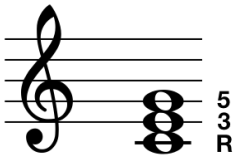
When do we use which number?! When do we say 2 and when is it 9, or a 6 instead of a 13? Is there a rule? Yes, there is. I've often taught people who have confusion over this topic so I'm going to try to clear it up from my understanding.

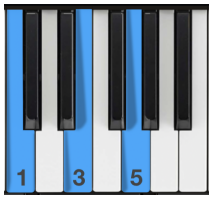
In music theory we like to put numbers on things like intervals, chords and chord tones.
Let's talk about chord tones. Chord tones are just the notes in our chords. In our basic triads we have the Root, 3 and 5. Any chord at its most basic and fundamental level is the root, 3rd and 5th. When all we have is the root, 3 and 5 we don't need to add anything else to our chord symbols.
But beyond that we start to learn about some more exotic or interesting chords and these have extra numbers added after our usual chord symbols.

Cmaj9 - C13 - Cm11...

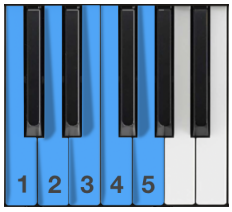
Now These extra numbers are just instructions or indicators of what extra notes to add to our original root 3rd and 5th or any alterations we might make to those notes. At the risk of stating the obvious, those extra numbers are referencing chord tones.

Let's take a closer look at our chord tones and how we number those.
Let's use a C major chord. I'm confident that we're all clear on the fact that C is our rootnote, or tonic, or we could call it the '1', although we don't typically, it's usually the root or tonic. Our E is the 3 or 3rd - by the way, these are interchangeable, 3 or 3rd - 5 or 5th. You're free to add the ordinal indicators should you wish.
So the C and E are our Root and 3rd and our G is the 5 of a C chord. And that makes up the fundamental triad of our chord.

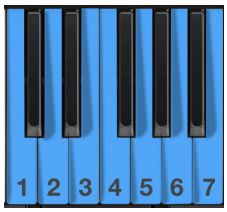




But, as you might guess, we could fill in those gaps with a 2 and a 4.

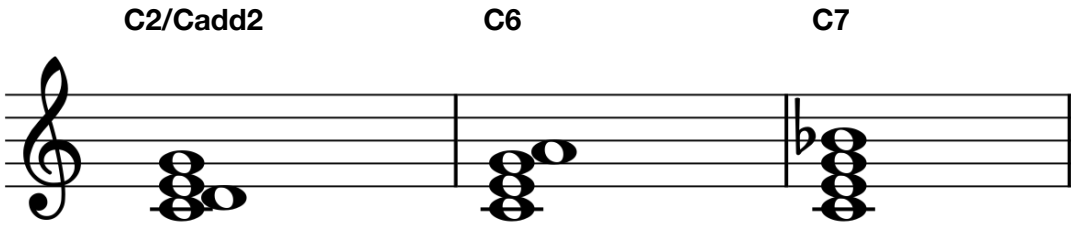


Why not let's go the whole hog and fill in the rest of the octave!

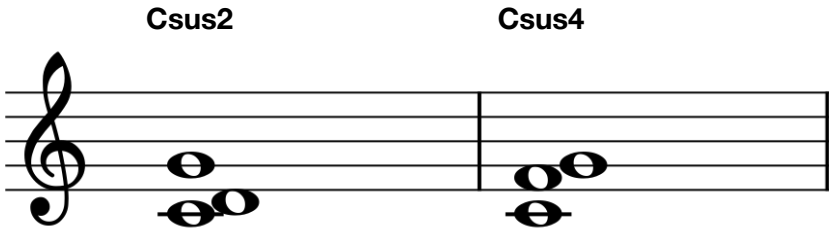


Ok so all the notes of the key have a number assigned, notes 1 to 7. And…those notes outside our key, or major scale, will either be a sharp or a flat of our neighbouring number.

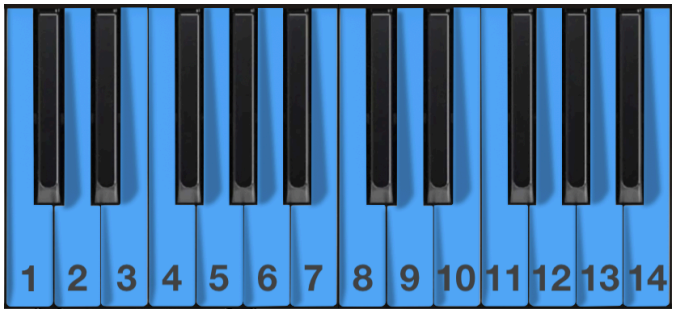
You've probably come across a host of chord symbols that include some of these extra notes: C7, Cadd2, Csus4, C6
Some of them simply add the note to our existing triad and some of them swap out notes from our triad for the other notes.
So for example we might see a C2, a C6 or a C7 and all those instances all we do is add that indicated chord tone to our existing triad. Sometimes you'll see C2 written as a Cadd2, they mean the same thing.



But unlike those we might sometimes come across a Csus2 or Csus4 and in this case we're actually swapping out one of the notes of our triad for this new chord tone. The 'sus' in a sus chord is short for suspended. In these chords we suspend the 3rd of the chord and replace it with either the 2 or the 4.



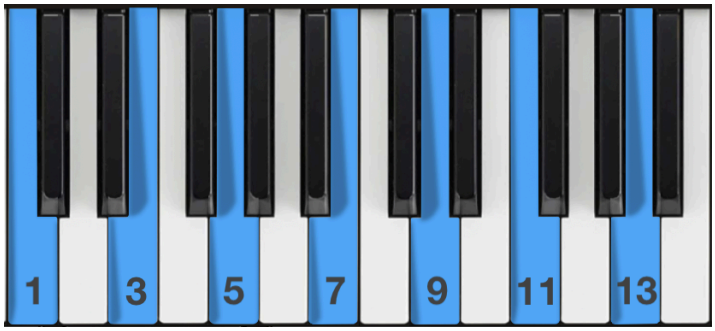
That stuff's fairly simple, now to the fun bit.
It's common to see chord symbols with 9s 11s and 13s. Well how is that possible when we've covered all the notes?! Well there is some sense to this apparent nonsense and it kind of comes back to how we create chords.
So what's obviously happening is that we're now counting above the octave.



So now we have two different numbers for each note. Our 2 and 9 are both D, our 6 and 13 are both A.
So is there a difference between a C6 chord and a C13 chord? Both of them would appear to add an A to our C chord. Well yes, there is a difference and I'll come on to that, but let's talk quickly about how we build chords.

BUILDING CHORDS

If we look back at our C major triad, this chord is built by taking our root note and adding notes in intervals of thirds, just using the notes from our key, another way of thinking about it is skipping a note each time. So with our simple triads we do this and stop at the 5th of the chord but we can keep going to extend our chords. So first we'd get our 7 then 9 then 11 and then 13 before we're done. If we did it again we'd be back to our 1, or root.



Now while we're here maybe it's worth saying that we never see 8, 10, 12 or 14 because they're already chord tones that we've got from the previous octave. Those notes are already accounted for in our chord.

So what is the difference? Here's the long and short of it:
Any chord that has numbers higher than 7 is indicating that you'd also include the chord tones beneath that number.

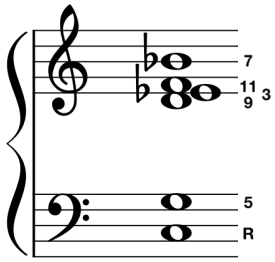
So a C9 chord not only includes our root, 3rd, 5th and 9th but also the 7th. We can think of the chord backwards or downwards from the 9 to the root - 9 7 5 3 1
So a C9 = CEGBb and D and alternatively, as I mentioned earlier, a C2 (or Cadd2) is merely adding that D to our triad - CDEG

A Cm11 chord has our minor triad plus our 7th 9th and 11th. Or, again, we could have worked backwards from our 11th - 11 9 7 5 3 1

There is a slight caveat with 13 chords and major 13 chords, they work the same way although we typically omit the 11 because the 11 alongside a major 3rd creates a dissonant interval of a minor 9th which spoils the colour of the chord. So a C13 chord includes C(R), E(3rd), G(5th), Bb(7th), D(9th) and A(13th) - No 11th (F).

COMMON MISUNDERSTANDING

A common misunderstanding or assumption that I come across from my students is that the reason we call it one number over the other is because it has something to do with what inversion you're playing or more specifically the octave. For example I've had people who've assumed the difference between a C6 and C13 is the position of the A, if it's within the octave it's a 6 and if it's above the octave here then it's a 13.
Now, that's not the case, the octave of the note in question does not affect what we call it. As I'm sure you're aware, inversions are a fundamentally important aspect of harmony, so we're always moving around the positions of our notes within octaves and never compromises the integrity of the chord or the chord name.
For example, I often play a voicing of the aforementioned Cm11 like this:



As you can see here I have my 11 and 9 in there with my 3 and 5. So it has nothing to do with what octave or inversion you're playing.

ONE LAST THING WORTH MENTIONING

Now there is one other thing that I've not mentioned that doesn't quite fall into what I've explained here. And that's a chord that you'll sometimes see which is an add9 chord.
My take on this is that it's exactly the same as an add2 and is maybe mistakenly and potentially confusingly called add9 when it should be add2.
I think this connects to something I think is a common occurrence in modern music theory. If you've got a second I'll tell you my thoughts. Nearly all modern music, and by that I mean popular music that's been created in the last hundred years - jazz, blues, rock, pop, funk, - all that. Nearly all that music has been learned either by word of mouth, from someone showing you how they do it, by listening and copying other people and learning in a way that makes sense to that person and they then teach others in a way that makes sense to them or how they like to think of it or understand it.
Unlike classical music which has a fairly established music theory history that has generally been agreed on for hundreds of years. Chord symbols are a pretty recent addition to music in the grand scheme of things and I think because of the way we've learnt modern music it's meant that we've ended up with variations of the same thing because people understand the theory from a different perspective and therefore might sometimes teach and communicate what we mean slightly differently from someone else. Most things in modern music have been absorbed into a conformed, agreed upon way of doing things, but I think chord symbols are something that we still commonly see variations of.

Getting back to my initial point. I think the general, agreed upon convention for chord naming and chord symbols is what I've laid out already but sometimes we see something like an add9 chord, which really should be an add2 chord, and doesn't fit neatly in our way of thinking. So effectively, if you see a chord tone with an 'add' before, it simply means we're adding that note to our triad and not anything else beneath it.

And…we'll leave it there. I know there are plenty of people who have this question about this subject so hopefully you're one of those people and have found this to be useful!

Thanks for being here!

MK